The Multiplication Rule KEY

Name _____

You guess on 3 multiple-choice questions. Each has 4 choices. What is the probability you get all 3 right? Each guess has a $\frac{1}{4}$ chance. The questions do not affect each other, so multiply:

$$P(\text{all 3 right}) = \frac{1}{4} \cdot \frac{1}{4} \cdot \frac{1}{4} = \frac{1}{64}$$

Independent events. Two events are **independent** if one happening does not change the probability of the other.

Independent: $P(A \cap B) = P(A) \cdot P(B)$

Example: Flip a coin and roll a number cube. $P(\text{heads} \cap 6) = \frac{1}{2} \cdot \frac{1}{6} = \frac{1}{12}$

Dependent events. Two events are **dependent** if one happening changes the probability of the other. A bag has 5 red and 3 blue marbles. Draw 2 marbles.

With replacement (put the first one back)	**Without replacement** (keep the first one out)
The second draw is the same as the first. Independent.	The bag changes: 4 red and 3 blue are left. Dependent.
$P(\text{red, red}) = \frac{5}{8} \cdot \frac{5}{8} = \frac{25}{64}$	$P(\text{red, red}) = \frac{5}{8} \cdot \frac{4}{7} = \frac{20}{56} = \frac{5}{14}$

General multiplication rule: $P(A \cap B) = P(A) \cdot P(B \mid A)$ $P(B \mid A)$ is read "the probability of B **given** A." It is the probability of B after A has already happened.

In the marble example: $P(\text{red second} \mid \text{red first}) = \frac{4}{7}$.

Tip: "At least one" questions are often easier with the complement.
$P(\text{at least one wrong}) = 1 - P(\text{all right})$.

Work with a partner. Compare answers when you finish each problem.

1. These events are independent. Find each probability.

   (a) Flip a coin 3 times. P(3 heads) $\frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8}$

   (b) Roll a number cube twice. P(two 6s) $\frac{1}{6} \cdot \frac{1}{6} = \frac{1}{36}$

   (c) A basketball player makes 80% of her free throws. P(she makes 2 in a row) $0.8 \cdot 0.8 = 0.64$

   (d) Same player. P(she misses both) $0.2 \cdot 0.2 = 0.04$

2. You guess on a 5-question true-or-false quiz.

   (a) P(all 5 right) $\left(\frac{1}{2}\right)^5 = \frac{1}{32}$

   (b) P(at least one wrong) $1 - \frac{1}{32} = \frac{31}{32}$

3. A bag has 4 green and 6 yellow marbles. Draw 2 marbles.

   (a) Find P(green, green) **with** replacement.
   $\frac{4}{10} \cdot \frac{4}{10} = \frac{16}{100} = \frac{4}{25}$

   (b) Find P(green, green) **without** replacement.
   $\frac{4}{10} \cdot \frac{3}{9} = \frac{12}{90} = \frac{2}{15}$

   (c) Why is the answer to (b) smaller?
   After a green is taken out, fewer greens are left, so the second green is less likely.

4. Independent or dependent? Explain.

   (a) Flip a coin, then roll a number cube.
   Independent. The coin does not change the cube.

   (b) Pick a class president, then pick a vice president from the students who are left.
   Dependent. The president cannot be picked again, so the choices change.

   (c) Draw a card, put it back, shuffle, and draw again.
   Independent. The deck is the same for the second draw.

   (d) It rains today, and you carry an umbrella today.
   Dependent. Rain makes carrying an umbrella more likely.

5. At a school, $P(\text{late}) = 0.2$ and $P(\text{detention} \mid \text{late}) = 0.5$.

   (a) Find $P(\text{late} \cap \text{detention})$. $0.2 \cdot 0.5 = 0.1$

   (b) Explain what your answer means in words.
   About 10% of all students are late and get a detention.